

Dichang Zhang

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Research Interests

Machine learning applications in physical science, including multi-fidelity fluid dynamics modeling and multi-survey astronomical image analysis.

Education

Stony Brook University, New York, United States

Ph.D. Candidate in Computer Science 2022–Present
Advisor: Prof. Dimitris Samaras

Ecole Polytechnique Federale de Lausanne, Lausanne, Switzerland

Visiting Doctoral Student in Computer and Communication Sciences 2025
Host advisor: Prof. Pascal Fua

University of Michigan, Ann Arbor, Michigan, United States

B.S. in Data Science and Honors Computer Science 2019–2022
Advisor: Prof. David Fouhey

Research Experience

Research Assistant, Stony Brook CV Lab, Stony Brook University

May 2023–Present
Faculty advisor: Prof. Dimitris Samaras

Visiting Researcher, Computer Vision Lab, Ecole Polytechnique Federale de Lausanne

January 2025–June 2025
Supervisor: Prof. Pascal Fua

Research Assistant, Fouhey AI Lab, University of Michigan

May 2020–March 2022
Faculty advisor: Prof. David Fouhey

Summer Research Assistant, State Key Lab of Fire Science, University of Science and Technology of China

June 2018–August 2018
Faculty advisor: Prof. Yuan Hu

Internship Experience

Research Intern, SIGMA Group, Futurewei Technologies

February 2026–August 2026
Mentor: Dr. Zhiqiang Lao

Publications

1. **Dichang Zhang**, Yixuan Shao, Jiali Cui, Haotian Yin, Yuanpeng Liu, Jiaqi Deng, Xiang Gao, Zhiqiang Lao, Heather Yu, Simon Birrer, Dimitris Samaras. “COSMIO: A Benchmark for Cross-Survey Modality Imputation.” *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. Accepted for publication.
2. Xiang Gao, Xiaoping Zhu, Xinmu Wang, Yuanpeng Liu, Haotian Yin, **Dichang Zhang**, Yazheng Chen, Yue Wang, Yubin Zhou, Yu Guo, Wei Chen, David Gu, Xinyu Song, Heather Yu. “Diffeomorphism-Informed 3D Gaussian Splattings via Screen-Space Optimal Transport (OT).” *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. Accepted for publication.
3. **Dichang Zhang**, Jiaqi Deng, Yixuan Shao, Yuanpeng Liu, Jiali Cui, Zhiqiang Lao, Heather Yu, Liang Peng, Simon Birrer, Dimitris Samaras. “A General-Purpose VLM Can Teach an Astronomy Foundation Model to Better Recognize Galaxy Morphology.” *arXiv:2608.02300 [cs.CV]*, 2026.
4. **Dichang Zhang**, Yixuan Shao, Simon Birrer, Dimitris Samaras. “AS-Bridge: A Bidirectional Generative Framework Bridging Next-Generation Astronomical Surveys.” *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2026.
5. Shilin Hu, Ke Ma, Sagnik Das, **Dichang Zhang**, Dimitris Samaras. “Deep Learning-Based Rail Surface Condition Evaluation.” *International Conference on Computer Vision (ICCV) Workshops*, 2025.
6. **Dichang Zhang**, Christian Santoni, Zexia Zhang, Dimitris Samaras, Ali Khosronejad. “Adaptive Multitask Neural Network for High-Fidelity Wake Flow Modeling of Wind Farms.” *Energies*, 2025.
7. Christian Santoni, **Dichang Zhang**, Zexia Zhang, Dimitris Samaras, Fotis Sotiropoulos, Ali Khosronejad. “Toward Ultra-Efficient High-Fidelity Predictions of Wind Turbine Wakes: Augmenting the Accuracy of Engineering Models via LES-Trained Machine Learning.” *Physics of Fluids*, 2024.
8. **Dichang Zhang**, Zexia Zhang, Christian Santoni, Ali Khosronejad, Dimitris Samaras. “Transfer Learning in Multi-Fidelity Surrogate Modeling: A Wind Farm Case.” *International Conference on Machine Learning (ICML) Workshops*, 2024.
9. Richard E. L. Higgins, David F. Fouhey, **Dichang Zhang**, Spiro K. Antiochos, Graham Barnes, J. Todd Hoeksema, K. D. Leka, Yang Liu, Peter W. Schuck, Tamas I. Gombosi. “Fast and Accurate Emulation of the SDO/HMI Stokes Inversion with Uncertainty Quantification.” *The Astrophysical Journal*, 2021.
10. Junling Wang, **Dichang Zhang**, Yan Zhang, Wei Cai, Congxue Yao, Yuan Hu, Weizhao Hu. “Construction of Multifunctional Boron Nitride Nanosheet Towards Reducing Toxic Volatiles (CO and HCN) Generation and Fire Hazard of Thermoplastic Polyurethane.” *Journal of Hazardous Materials*, 362:482–494, 2019.
11. Wei Cai, **Dichang Zhang**, Bo Wang, Yuxuan Shi, Ying Pan, Jin Wang, Wen Hu, Yuan Hu. “Scalable One-Step Synthesis of Hydroxylated Boron Nitride Nanosheets for Obtaining Multifunctional Polyvinyl Alcohol Nanocomposite Films: Multi-Azimuth Properties Improvement.” *Composites Science and Technology*, 2018.
12. Feng Chu, **Dichang Zhang**, Yifan Hou, Shiyu Qiu, Jin Wang, Wen Hu, Lei Song. “Construction of Hierarchical Natural Fabric Surface Structure Based on Two-Dimensional Boron Nitride Nanosheets and Its Application for Preparing Biobased Toughened Unsaturated Polyester Resin Composites.” *ACS Applied Materials & Interfaces*, 2018.